

Engineering Ethics

“Risks, Accidents and Liability”

Hochschule Darmstadt University of Applied Sciences

May 15, 2022

Agenda for today

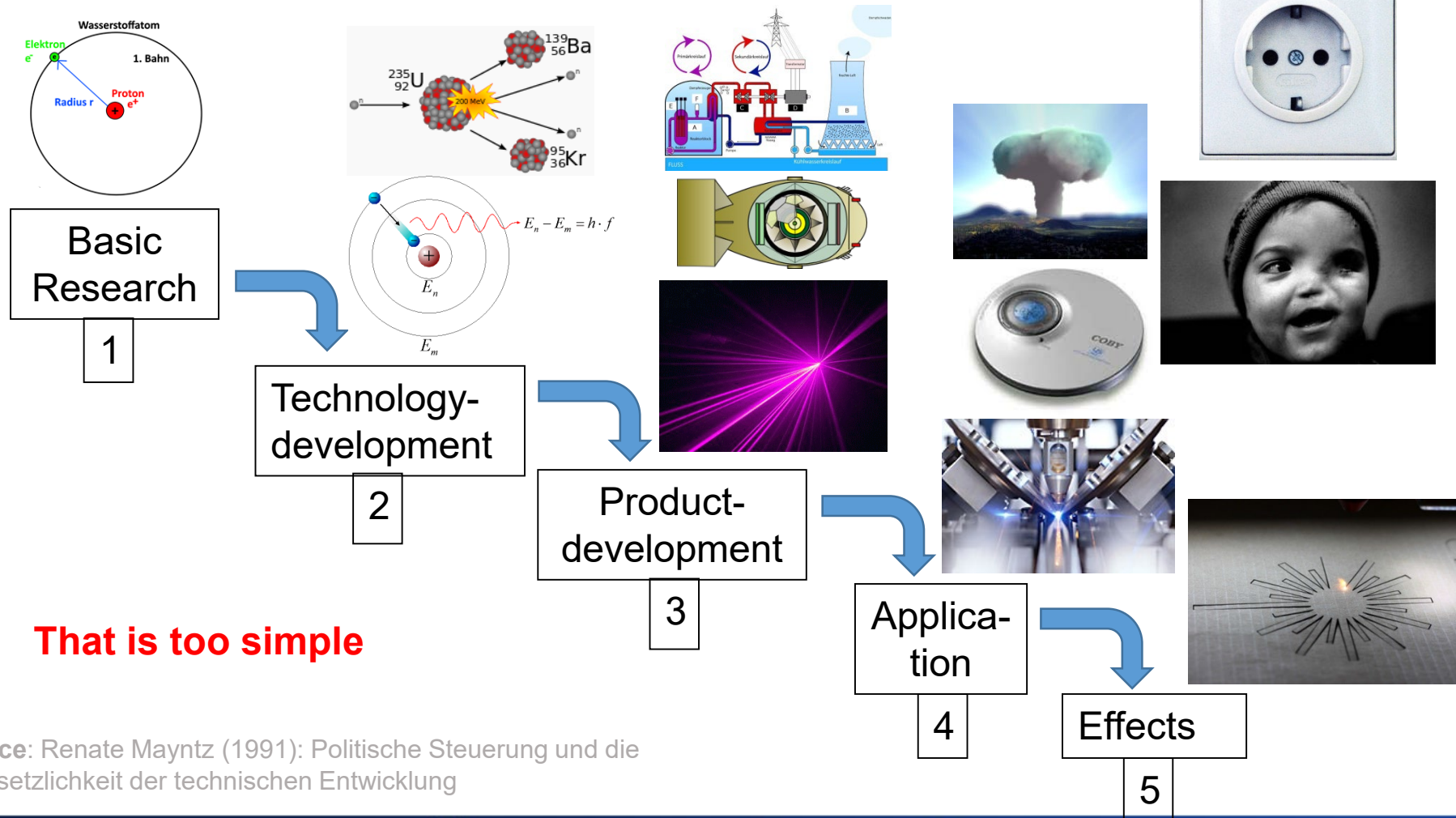
1. Introduction
2. A Short Recapitulation
3. An Example From Everyday Life
4. Living With Risks
5. Risk Evaluation
6. Case No. 1

Introduction

- Any questions?

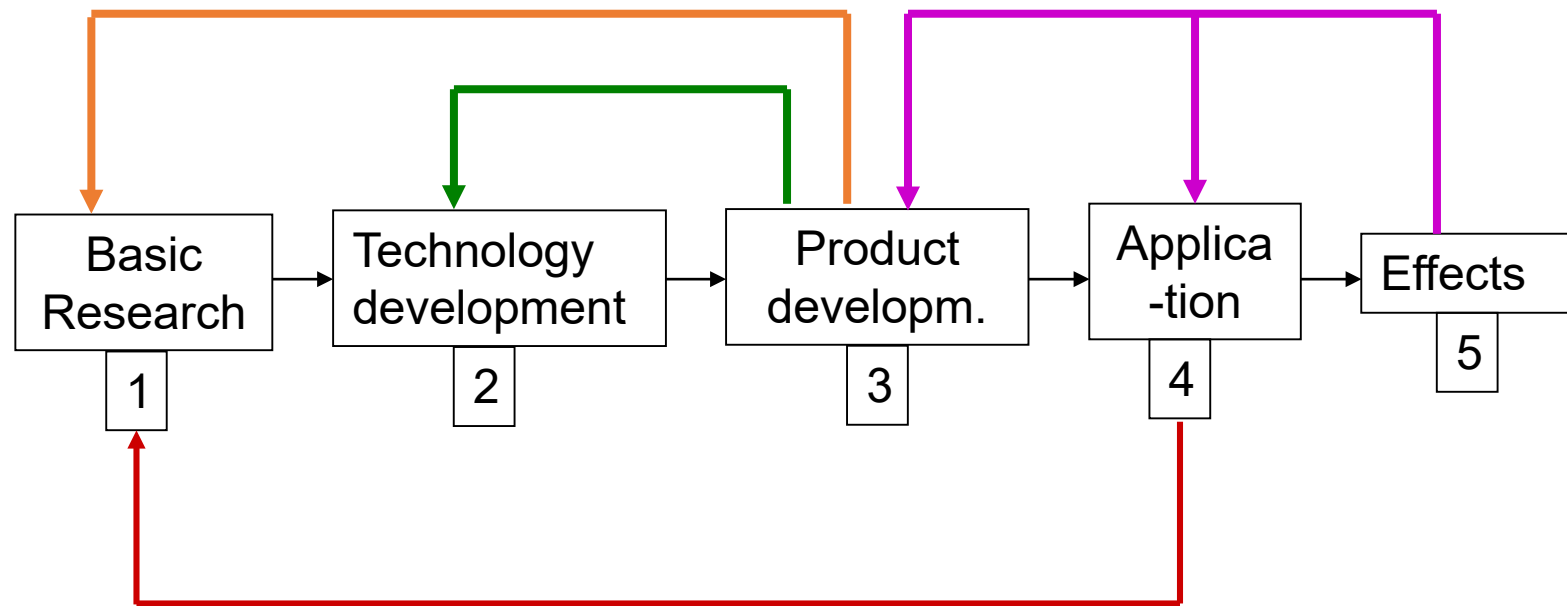
1

A phase scheme of technical development



2

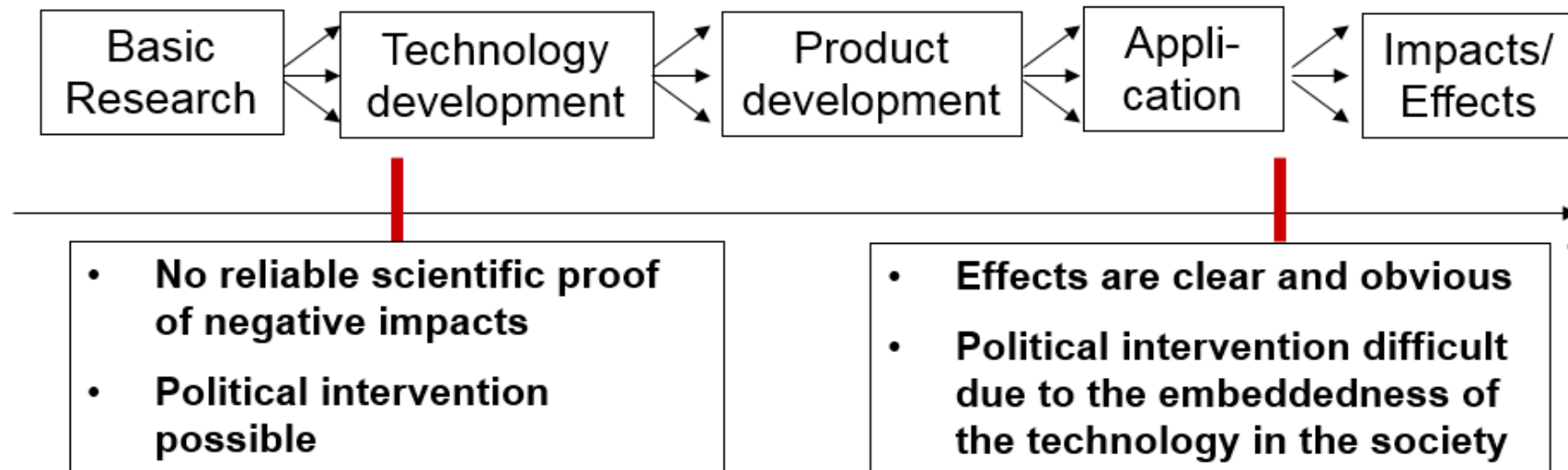
Reference: Renate Mayntz (1991): Politische Steuerung und die Eigengesetzlichkeit der technischen Entwicklung



Example:
Laser material processing



The Collingridge Dilemma



Collingridge, D. (1980). The social control of technology. St. Martin. New York.

Consequences and effects

Consequences are



Chances and **Risks** !!!



Discussion arise typically when it comes to the question of how (equally) **Chances** and **Risks** are distributed

An example from everyday life!

3



A person

...



Taken from
Perrow, C. (2011 (first 1984). *Normal accidents: Living with high risk technologies-Updated edition*. Princeton university press.

... has a job interview in
a big city at 10:00 am.



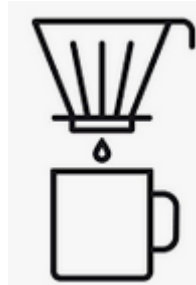
A person



The coffee machine



This was the essential part



Redundant solutions



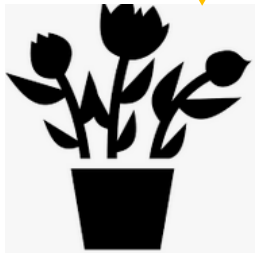
Sometimes time consuming



Where are my keys?



Inside! And the door is locked



A spare key in the flower pot



"I gave it to a neighbour yesterday!"



Now you're really in a hurry



It is good to have another nice neighbour with a car



„Unfortunately the car is in repair today“



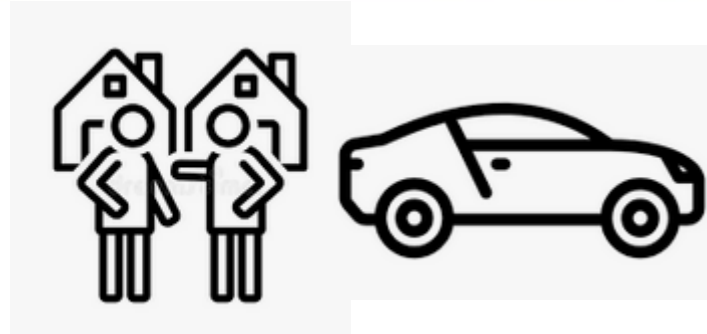
A spare key
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Now you're
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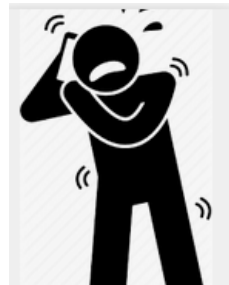
It is good to have a nice
neighbour with a car



„Unfortunately the
car is in repair today“



Bus drivers
are on strike today



TAXI !



„Because of the strike, all the taxis are
out and the streets are very crowded.
It may take a little longer!“



... has a job interview in
a big city at 10:00 am.

What was the primary cause of the „accident“?

- ☐ Human error (e.g. key inside of the house)
- ☐ Technical defect (e.g. neighbours car in repair)
- ☐ The environment (e.g. bus drivers on strike)
- ☐ System design (e.g. no emergency taxis in case of a strike)+
- ☐ Procedures used (e.g. keeping the coffee warm in the coffee maker)

Go to ...

www.menti.com

Use code:

9873 6342

What was the primary cause of the „accident“?

What was the primary cause of the „accident“?

- The best answer to the 5 questions in all 5 cases is: "No".
- There is no "primary" cause, this lies rather in the complexity of the circumstances.
- Each individual aspect in itself is trivial.
- Only the coincidental occurrence of all these trivial aspects together at one point in time led to the "catastrophe".

What was the primary cause of the „accident“?

- The cause of the incident lay in the interactive nature of the world and its sometimes tight couplings:
 - Most of the time we take no notice of these things because breakdowns are rare or because minor disruptions can be compensated for by "redundancy paths".
 - However, it can happen that things that at first glance seem to have no connection (bus, keys and coffee machine...) are linked together.

(Living with) Risks

4



<https://www.youtube.com/watch?v=MTdFkPTTnsA> (min. 1:30-2:20)

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(Living with) Risks

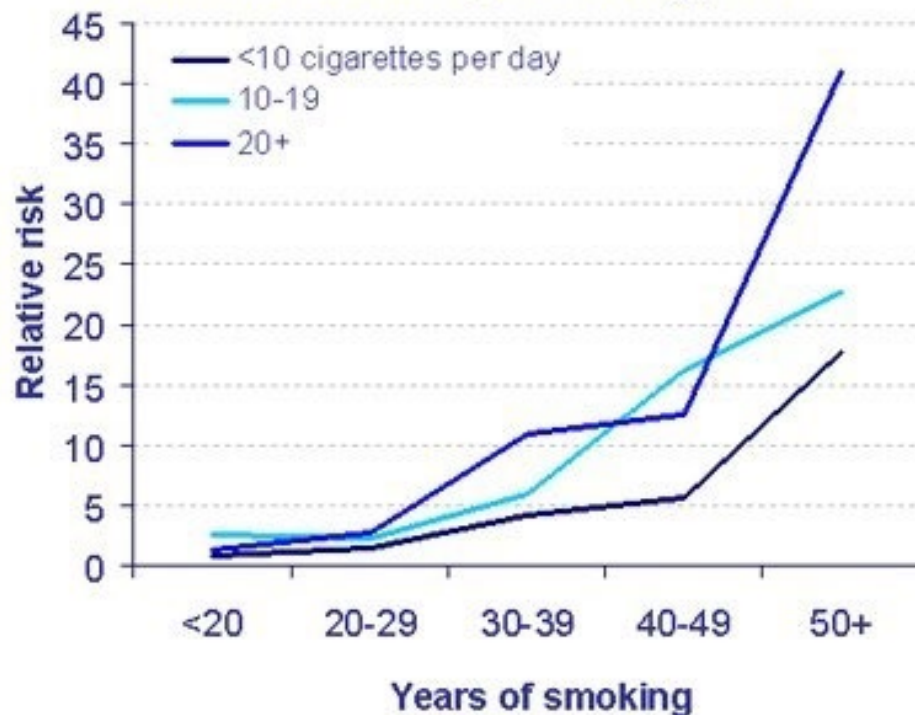
Everyday Risk Perception and Assessment



		CONSEQUENCES – WHAT IS THE MAXIMUM REASONABLE CONSEQUENCE				
		Insignificant	Minor	Moderate	Major	Catastrophic
LIKELIHOOD RATING	Almost certain	Medium	Medium	High	Extreme	Extreme
	Likely	Low	Medium	Medium	High	Extreme
	Possible	Low	Low	Medium	High	High
	Unlikely	Low	Low	Low	Medium	High
	Rare	Low	Low	Low	Low	Medium

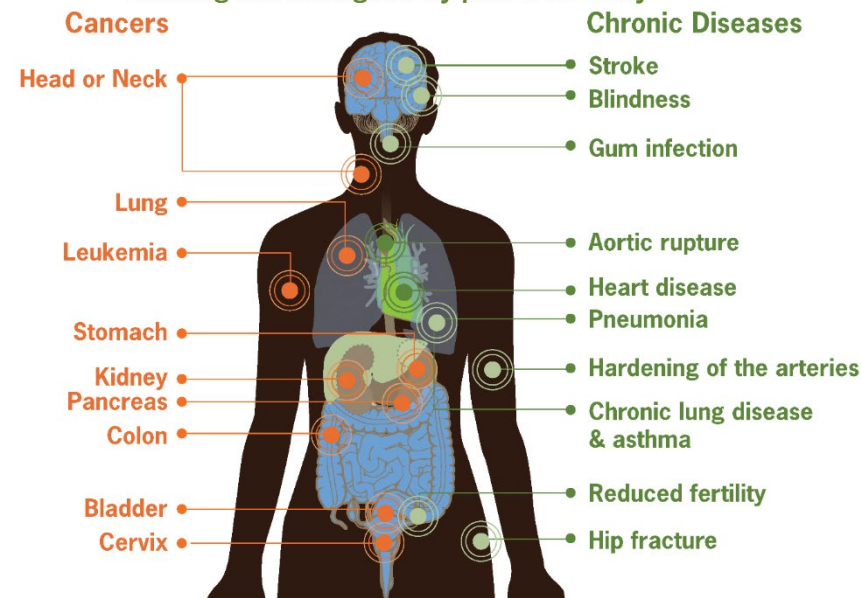
Everyday Risk Perception and Assessment

Figure 1.1: Relative risk of lung cancer, according to duration and intensity of smoking, men



Risks from Smoking

Smoking can damage every part of the body



<https://www.quora.com/How-much-numerically-does-tobacco-smoking-increase-your-lung-cancer-risk>

Hazard vs. Risk

- Hazard:
Something that has the potential to cause harm to people, property or the environment.
- Risk:
The potential that a hazard will cause harm – the chance or probability of that hazard causing harm or damage to people, property or the environment.



Annual Global Road Crash Statistics

- Approximately 1.35 million people die in road crashes each year, on average 3,700 people lose their lives every day on the roads.
- An additional 20-50 million suffer non-fatal injuries, often resulting in long-term disabilities.
- More than half of all road traffic deaths occur among vulnerable road users—pedestrians, cyclists, and motorcyclists.
- Road traffic injuries are the leading cause of death among young people aged 5-29. Young adults aged 15-44 account for more than half of all road deaths.
- More than 90% of all road fatalities occur in low- and middle-income countries, even though these countries have approximately 60% of the world's vehicles.

<https://www.asirt.org/safe-travel/road-safety-facts/>

Risk Management of Insurance Companies

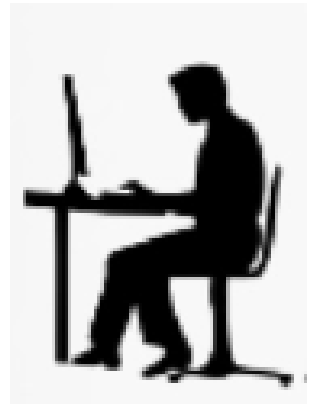
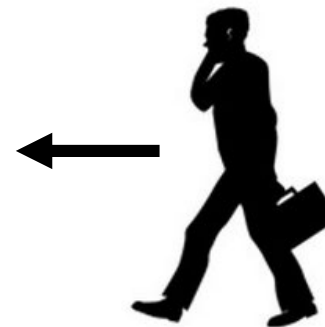
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Probability of a damage x cost of damage = risk

x% of 45 Mio cars in Germany cause a damage x Average cost per car accident = Total amount of damage

Overall insurance premium = + administrative costs and profit

Payments by insurance

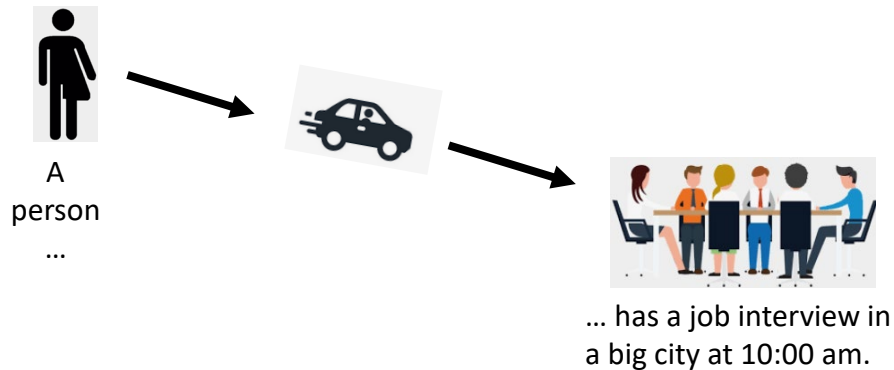


Risk evaluation

- Voluntary vs involuntary risks
- Short term vs. long term consequences
- Expected probability (1 in 1,000,000 severe vs. 50:50 chance minor impact)
- Reversible effects
- Threshold level for risks increase tolerance
- Delayed vs. immediate risk (high-fat diet vs. skydiving)

An example from everyday life!

6



Where is the
connection with
Engineering Ethics?

Charles Perrow – Analytical Tool

D → Design
E → Equipment
P → Procedures
O → Operators
S → Supply and Materials
E → Environment



These features come together
when technologies or
technical products are
developed and used.

The Eschede derailment



Read the case and apply
the characteristics

D → Design

E → Equipment

P → Procedures

O → Operators

S → Supply and Materials

E → Environment

- You have about 20 Minutes
- Discuss the question in your group and
- Write your thoughts down
- Write your names on the paper